An Introduction to the RSA Encryption Method

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Outline

1. History
2. Modular Arithmetic
3. Your own RSA system
4. Example
5. Proof

History

- RSA stands for Rivest, Shamir, and Adelman, the last names of the designers
- It was first published in 1978 as one of the first public-key crytographic systems
- A public-key system means the algorithm for encrypting a message is publicly known but the algorithm to decrypt the message is only privately known (by the person who set up the system)

Modular Arithmetic Review

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Necessary Theorems for RSA - $\phi(n)$

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Corollary

If $b_1 \equiv b_2 \pmod{\phi(n)}$, then $a^{b_1} \equiv a^{b_2} \pmod{n}$.

Setting up your own RSA system

- Pick p and q to be large prime numbers, and let $n = pq$. Then pick an e such that $gcd(e, \phi(n)) = 1$. e is your encryption exponent.

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- You now have your own RSA system!
  - Public Key - (n, e)
  - Private Key - (d)

Using your RSA system

When someone wants to send you a message they:

1. Convert their message into a number in a simple agreed upon way such as a=01, b=02, c=03 ...
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To decrypt their message you:

1. Compute $m \equiv c^d \pmod{n}$
2. Convert their message back into letters and words

Example (Set-Up and Encryption)

First, set up your RSA system.

- Pick $p = 5, q = 11$. Let $n = pq = 55$. Now pick $e = 3$.
- Then $ed \equiv 1 \pmod{\phi(n)} \implies d = 27$. Since $3 * 27 \equiv 81 \equiv 1 \pmod{40}$.
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Let's say that your friend wants to send you the message m=18.

- They will compute c where $c \equiv m^e \pmod{n}$. $c \equiv m^e \equiv 18^3 \equiv 5832 \equiv 2 \pmod{55}$ because $18^3 = 5832 = 2 + (106)55$.
- Your friend will send you the ciphertext $c = 2$.

Example (Decryption)

You just recieved $c = 2$ from your friend.

- Use your private key, $d = 27$, to compute their message m. $m \equiv c^d \equiv 2^{27} \equiv 134217728 \equiv 18 \pmod{55}$ because $2^{27} = 134217728 = 18 + (2440322)55$.
- So your friend sent you the message $m = 18$.

RSA Proof

Why does $m \equiv c^d \pmod{n}$ work to get you back the original message m?

Proof.

Let p and q be prime, $n = pq$, $ed \equiv 1 \pmod{\phi(n)}$. Then $\exists k \in Z$ such that $ed = 1 + k\phi(n)$.
Also let $m < n$ be a message and let $c \equiv m^e \pmod{n}$. Then, $c^d \equiv (m^e)^d \equiv m^{ed} \equiv m^{1+k\phi(n)} \equiv m \pmod{n}$. □